

Ex: 11 SIMPLE TOPOLOGY AND CONFIGURE WITH ONE ROUTER, TWO SWITCHES AND PCS

AIM

To design and configure a simple network topology in Cisco Packet Tracer using one router, two switches, and four PCs. The goal is to enable communication between PCs across different networks via the router.

DEVICES REQUIRED

Device	Type	Quantity	Model Suggestion
Router		1	Cisco 2911 or 1941
Switch		2	Cisco 2960
PC		4	Generic PC
Cables		6	Copper Straight-Through

CONFIGURATION STEPS

1. Physical Connections

- Connect PC0 and PC1 to Switch0 (FastEthernet0/1 and 0/2).
- Connect PC2 and PC3 to Switch1 (FastEthernet0/1 and 0/2).
- Connect Switch0 to Router0 via GigabitEthernet0/0.
- Connect Switch1 to Router0 via GigabitEthernet0/1.

2. IP Addressing

Assign IPs and subnet masks:

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	Gig0/0	192.168.1.1	255.255.255.0	—
Router0	Gig0/1	192.168.2.1	255.255.255.0	—
PC0	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.1

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC2	FastEthernet0	192.168.2.10	255.255.255.0	192.168.2.1
PC3	FastEthernet0	192.168.2.11	255.255.255.0	192.168.2.1

3. Router Configuration

enable

configure terminal

interface gig0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gig0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

end

4. PC Configuration

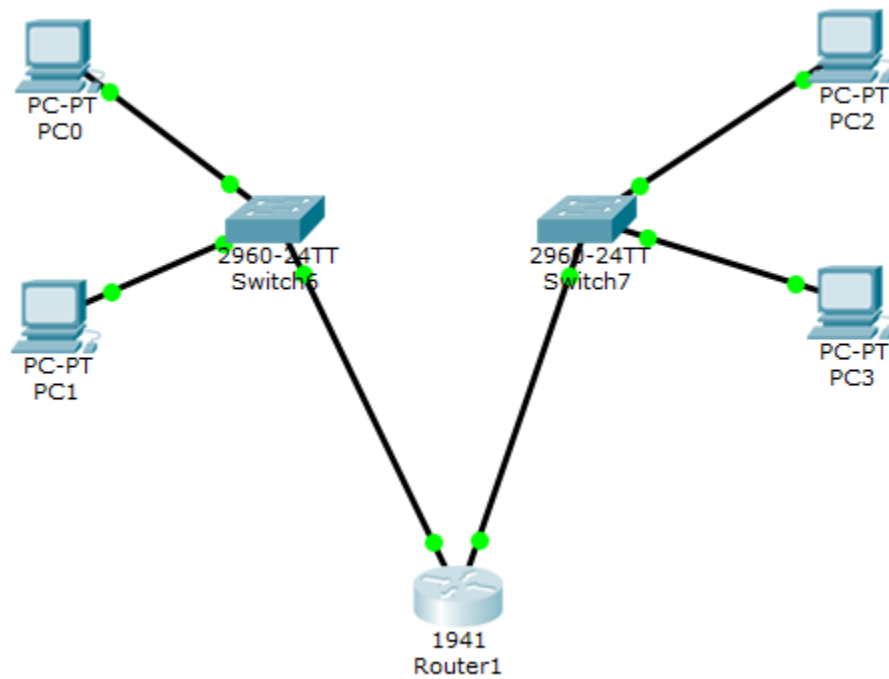
Go to each PC → Desktop → IP Configuration:

Set IP address, subnet mask, and default gateway as per the table above.

OUTPUT:

Once configured:

- PCs within the same network (e.g., PC0 ↔ PC1) can ping each other.
- PCs across networks (e.g., PC0 ↔ PC2) can communicate via the router.
- Use the ping command in Command Prompt on PC0:



RESULT:

This setup ensures that you have a simple network topology with one router, two switches, and four PCs, with communication between the PCs across different networks. The router provides the necessary inter-network routing, allowing all PCs to communicate with each other.